Lec-1

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Java divides the into the following groups:

- Arithmetic operators
- Assignment operators
- Comparison operators
- Logical operators
- Bitwise operators → Bit-masking

Arithmetic Operators

Arithmetic operators are used to perform common mathematical operations.

Operator	Name	Description	Example	Try it
+	Addition	Adds together two values	x + y	Try it »
-	Subtraction	Subtracts one value from another	x - y	Try it »
*	Multiplication	Multiplies two values	x * y	Try it »
/	Division	Divides one value by another	x / y	Try it »
%	Modulus	Returns the division remainder	x % y	Try it »
++	Increment	Increases the value of a variable by 1	++x	Try it »
--	Decrement	Decreases the value of a variable by 1	--x	Try it »

+ - * / %

++
--

ADVERTISEMENT

Java Assignment Operators

x = 5

x = = 5

Assignment operators are used to assign values to variables.
In the example below, we use the **assignment** operator (=) to assign the value **10** to a variable called **x**:

Example

```
intx =10;
```

Try it Yourself »

The **addition assignment** operator (+=) adds a value to a variable:

Example

```
intx =10;x +=5;
```

Try it Yourself »

A list of all assignment operators:

Operator	Example	Same As	Try it
=	x = 5	x = 5	Try it »
+=	x += 3	x = x + 3	Try it »
-=	x -= 3	x = x - 3	Try it »
*=	x *= 3	x = x * 3	Try it »
/=	x /= 3	x = x / 3	Try it »
%=	x %= 3	x = x % 3	Try it »
&=	x &= 3	x = x & 3	Try it »
|=	x |= 3	x = x | 3	Try it »
^=	x ^= 3	x = x ^ 3	Try it »
>>=	x >>= 3	x = x >> 3	Try it »
<<=	x <<= 3	x = x << 3	Try it »

x+=3

x= x+3

~~x+=~~

x -=3

x=x-3

Java Comparison Operators

Comparison operators are used to compare two values:

Operator	Name	Example	Try it
==	Equal to	x == y	Try it »
!=	Not equal	x != y	Try it »
>	Greater than	x > y	Try it »
<	Less than	x < y	Try it »
>=	Greater than or equal to	x >= y	Try it »
<=	Less than or equal to	x <= y	Try it »

x = = y

x != y

x >= y | x > y

x <= y | x < y

Java Logical Operators

Logical operators are used to determine the logic between variables or values:

Operator	Name	Description	Example	Try it
&&	Logical and	Returns true if both statements are true	x < 5 && x < 10	Try it »
||	Logical or	Returns true if one of the statements is true	x < 5 || x < 4	Try it »
!	Logical not	Reverse the result, returns false if the result is true		

data TYpes

Type ← x

int x= 5

5

5.9

logical

and or not

a	b	a && b	a||b
T	F	F	T
F	T	F	T
F	F	F	F
T	T	T	T

~(T) = F

~(F) = T

!(T) = F

!(F) = T

P, R T 8

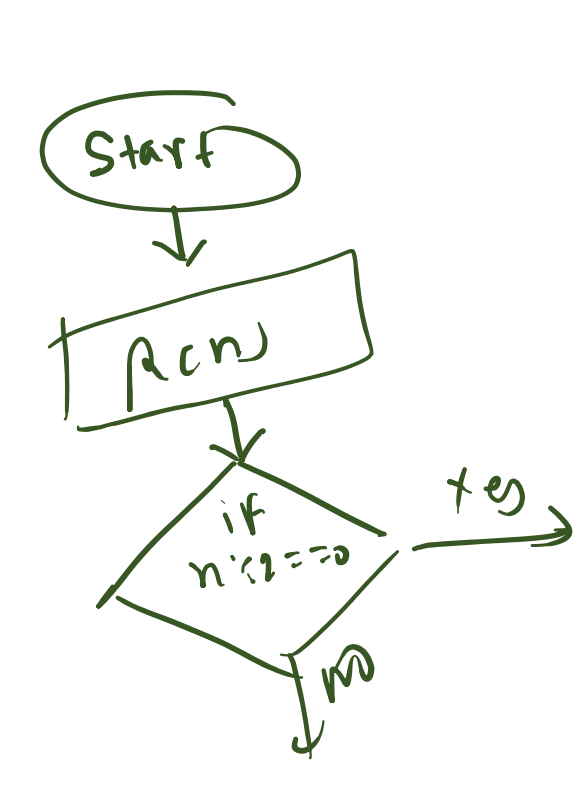
1000 1

simple

① n→ n%2 = = 0 → even

↳ odd

%n = 0, 1 to n-1



For →

while

do while →

For each →

n= 5 i=1

while (i ≤ n) {

i++

}

```
int n = 5;
    int i = 1;
    while (i <= n) {
        System.out.println("Hello");// self work
        i = i + 1;
    }
```

1 <= 5

i=1	1<=5
i=2	2<=5
i=3	3<=5
i=4	4<=5
i=5	5<=5
i=6	6<=5 X

Hello
Hello
Hello
Hello
Hello

```
 int x = 8;
int c = x++ + --x + ++x + x--;
System.out.println(c);
```

x = ~~8~~ ~~9~~ ~~8~~ ~~9~~ 8

8 + 8 + 9 + 9

```
Ques 1:
n = 5

*   *   *   *   *
*   *   *   *   *
*   *   *   *   *
*   *   *   *   *
*   *   *   *   *
```

N = ~~5~~ ~~10~~ 100

* * * * *

```
Ques 1:
n = 5

*   *   *   *   *
*   *   *   *   *
*   *   *   *   *
*   *   *   *   *
*   *   *   *   *
```

Row =1 Star=N.

while (Row ≤ n) {

//Star

int i=1

while (i <= Star) {

SYSo('*')

i++

}

//next line prep

Row++

SYStem.out.println()

}

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    int row = 1;
    int star = n;
    while (row <= n) {
        // star
        int i=1;
        while(i<=star) {
            System.out.print("* ");
            i++;
        }
        // next line ki prep
        row++;
        System.out.println();
    }

}
```

1<= 5, 2<=5, 3<=5 4<=5, 5≤5

Row = ~~1~~ ~~2~~ ~~3~~ ~~4~~ 5 6 Star= 5

1 <= 5

```
*   *   *   *   *
*   *   *   *   *
*   *   *   *   *
*   *   *   *   *
*   *   *   *   *
```

```
Ques 2:
n = 5

*
*   *
*   *   *
*   *   *   *
*   *   *   *   *
```

Row=1 Star= 1

while (Row≤n) {

//Star

//nextline

Row++

SYSo()

Star++

}

```
Ques 4:
n = 5

                *
            *   *
        *   *   *
    *   *   *   *
*   *   *   *   *
```

5

Row=1 Space = N-1 Star = 1

while (Row ≤ n) {

//Space

int i=1

While (i <= Space) {

System.out.Print(" ");

i++

}

//star

int j=1

while (j <=Star) {

SYSO ("*")

j++

}

//next line Prep

SYSo()

Row++

Star++

Space

}

```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        int row = 1;
        int star = 1;
        int space = n - 1;
        while (row <= n) {
            // sapce
            int i = 1;
            while (i <= space) {
                System.out.print("  ");
                i++;
            }
            // star
            int j = 1;
            while (j <= star) {
                System.out.print("* ");
                j++;
            }
            // next line ki prep
            row++;
            System.out.println();
            star++;
            space--;
        }

    }
```

5≤5

1<=5 → 2<=5, 3<=5

0 ~~1~~ 2

4

Row = ~~1~~ ~~2~~ ~~3~~ ~~4~~ 5

Star = ~~1~~ ~~4~~ ~~3~~ ~~2~~ 5

Space = 4 ~~3~~ ~~2~~ ~~1~~ 0

```
- - - - *
- - - * *
- - * * *
- * * * *
* * * * *
```

```
Ques 6:
n = 5

*   *   *   *   *
    *   *   *   *
        *   *   *
            *   *
                *
```

0 2 4 6 8

Row=1 Space=0 star= N

while (Row ≤ n) {

// space

// star

// next line prep

space += 2

star --

Row ++

SYSo()

}

```
Ques 9:
n = 5

                *
            *   *   *
        *   *   *   *   *
    *   *   *   *   *   *   *
*   *   *   *   *   *   *   *   *
```

N = 5 Row=1 Star= 1 Space = N-1

while (Row ≤ n) {

// Space

// star

//nextline prep

Row++

SYSO

Star += 2

Space --

}

```
public static void main(String[] args) {
        int n = 5;
        int row = 1;
        int star = 1;
        int space = n - 1;
        while (row <= n) {
            // sapce
            int i = 1;
            while (i <= space) {
                System.out.print("  ");
                i++;
            }
            // star
            int j = 1;
            while (j <= star) {
                System.out.print("* ");
                j++;
            }
            // next line ki prep
            row++;
            System.out.println();
            star += 2;
            space--;
        }
    }
```

row = ~~1~~ ~~2~~ 3

Space = ~~4~~ ~~3~~ 2

2<= 5

```
Ques 12:
n = 5

                *
            *   !   *
        *   !   *   !   *
    *   !   *   !   *   !   *
*   !   *   !   *   !   *   !   *
```

N = 5 Star= 1 Space=N-1

Star = ~~1~~ 3 5

```
- - - - *
- - - * * *
- - * ! * ! !
```

j=1,3 * *

j=2 !

j=1 3 7

j=2 4